



Technical Background of the
Paddy Crop Cutting Survey in
Sri Lanka



Agriculture and Environment Statistics Division
Department of Census and Statistics
Democratic Socialist Republic of Sri Lanka

Technical Background of the Paddy Crop Cutting Survey in Sri Lanka

2022

K. W. S. Saddhananda

Director Statistics

Agriculture and Environment Statistics Division

Department of Census and Statistics

Democratic Socialist Republic of Sri Lanka

Foreword

Rice is the staple food of the vast majority of Sri Lankans and at least two meals from three main meals comprises with rice and curry. Therefore it is very necessary to estimate local rice production correctly on time annually. This important activity is done by the Department of Census and Statistics since 1952 continuously. It is organized as sample survey to cover all over the country in each cultivation season.

The decision making of the rice consumption of the community is taken based on the paddy productions which are estimated by this survey. But there is no correct information to the usage of rice for other food productions, beverages and animal feeds as well as post-harvest losses and wastages. Therefore the availability of local rice production for human consumptions is estimated based on some assumptions.

All the calculations are done by using the findings of the sample survey and the estimated production is a relative figure to the correct absolute production of the relevant cultivated season. However, the sampling techniques have been arranged to minimize the sampling errors and therefore it can be assumed that the estimated figures are comparatively very close to the absolute productions.

This important report is prepared to clarify the technical background of this sample survey and this will be guided to the future modifications of the survey.

My special thanks to the officers of the Department of Census and Statistics for their valuable contributions to the fulfillment of the report.

K. W. S. Saddhananda

Director Statistics

Agriculture and Environment Statistics Division

Department of Census and Statistics

306/71, Polduwa Rd.

Battaramulla

Sri Lanka.

2022.02.07

Contents

	Page No.
Foreword	v
1. Paddy Production in Sri Lanka	1
1.1 Historical Background	1
2. Estimation of the paddy production in Sri Lanka	2
2.1 Original entry	2
2.2. Paddy production and consumption in Sri Lanka	2
2.3. Paddy cultivation by administrative district in Sri Lanka	2
2.4. Mode of irrigation for cultivation	3
2.5 Elementary practices of the sampling design	5
2.51 Mathematical way of thinking and notations	6
3. Water supply for cultivation	8
3.1 Pattern of the monsoon winds and rains in Sri Lanka	8
3.2 Agro ecological regions in Sri lanka	8
3.3 Identification of climatic zones by divisional secretariats	9
4. Sample selection to the paddy crop cutting survey	11
4.1 Sample selection	11
4.2 Selection of the experimental paddy plots	12
4.3 Harvesting of the experimental paddy plots	15
4.4 Measuring the harvested yield	16
4.5 Record the result of the experiment with the other related information	17
5. Estimation of the paddy production	18
5.1 Estimation of the annual paddy production and rice requirements of consumers	18
5.2 Future planning and solutions to the crisis	19
 List of Tables	
Table 2.1 : Asweddumized area of paddy cultivation by Administrative districts in Sri Lanka – 2021	3
Table 3.1 : Cultivation seasons with monsoon winds by months of calendar year	8
Table 3.2 : Number of DS divisions to the climatic zones by administrative district	10
Table 4.1 : Bonus experiments to the DS division by climatic zone	11
Table 4.2 : Percentage distribution of proportional allocations to the DS division by climatic zones and monsoonseason	11

	Page
Table 4.3 : Experimental parcel sizes of paddy crop cutting survey	15
Table 4.4 : Sizes of the sheer set by metric units	17
Table 5.1 : Paddy production and consumption pattern in Sri Lanka - 2021	19
Table 5.2 : Asweddumized area of paddy cultivation and expected production by Administrative district in Sri Lanka – 2021	20
Table 5.3 : Asweddumized area of paddy cultivation and estimated production by Administrative district in wet zone in Sri Lanka – 2021	21
 List of Figures	
Figure 2.1 : Administrative Districts in Sri Lanka	4
Figure 2.2 : Divisional Secretariats in Anuradhapura Districts in Sri Lanka	4
Figure 2.3 : Irrigation modes in Horowpathana Divisional Secretariat in Anuradhapura District	5
Figure 3.1 : Agro-Ecological Regions in Sri Lanka	9
Figure 4.1 : Sketched lay out of the selected paddy field	13
Figure 4.2 : Sketched lay out of the selected discernible square area	14
Figure 4.3 : Reaping the rice plants manually	15
Figure 4.4 : Threshed the paddy manually by using the scaffold	16
Figure 4.5 : Removing husks, chaff and other impurities of the paddy	16
Figure 4.6 : The harvested paddy yield is ready to measure	17
 Appendix	22
CC1 form	23
CC2 form	24
CC3 form	25
CC4 form	27
CC5 form	28
Table A1 : Seed variety with average life cycle (Harvesting Period)	29

1. Paddy Production in Sri Lanka

1.1 Historical Background

Rice is the staple food of the inhabitants of Sri Lanka. Ancient and traditional farmers have completely used organic fertilizer for their cultivation and did not induce any harm to the surrounding environment.

Sri Lanka was self-sufficient in rice production at the ancient period and the country was among the foremost paddy exporters in the world on that period. Paddy production in the country had being flourished during the reigns of Sri Lankan kings and it became deteriorated gradually after foreign invasion after 1505 AD.

Patriotic intellectual Hon. D. S. Senanayake, who became the first prime minister of the independent Sri Lanka in 1948, entered the Legislative Council in 1922. Thereafter, he was elected to the State Council in 1932 and appointed minister for agriculture and lands. He could able to hold the post for 15 years. Within two years he had put through the Land Development Ordinance, which brought barren lands into cultivation through irrigation schemes. He has started the Gal Oya development project on that period. The Gal Oya river is the main source feeding the Gal Oya scheme, a government programme that dammed this river and other smaller rivers to create Senanayake Samudraya, the largest tank (reservoir) in Sri Lanka, at Bintenne. The project has opened up 100,000 acres (40,000 hectares) of land to the cultivation of paddy, sugarcane, chilies, and other crops throughout the eastern coast.

The Mahaweli Development program is known as the largest multipurpose national development programme in the history of Sri Lanka after independence and is also considered as the keystone of the government's development programme that was initiated in 1961. However, this project was virtually activated at the latter part of the 1970 decade as an accelerated Mahaweli Development Project with five major reservoirs namely Victoriya, Randenigal, Rantambe, Kothmale and Maduruoya.

Generation of hydroelectric power, controlling floods, making irrigation facilities for dry zone cultivation, settlement of landless and unemployed families by constructing and developing the physical and social infrastructure required for human habitation by using the waters of the Mahaweli River were the main purposes to carrying out this program. Increase local agricultural production and create employment opportunities for the people were among the other expected advantages of this project.

Udawalawa, Rambacan Oya, Moragahakanda and Kaluganga projects were also introduced by the same authority and successfully completed to provide various facilities to the agricultural community in the country.

2. Estimation of the paddy production in Sri Lanka

2.1 Original entry

The inception of the paddy crop cutting experiments was happened in 1952 and the Department of Census and Statistics in Sri Lanka is the responsible authority for this momentous task.

The paddy production was considerably increased with the Mahaweli Accelerated Development Multipurpose Programme and completion of other related agricultural projects. However, the usage of paddy production for the other food and beverages were also increased with the improvement of the industry sector under the open economy which was introduced in 1978.

2.2. Paddy production and consumption in Sri Lanka

Estimated net paddy production was 4,710,211 metric tons in the year 2021 and the availability of raw rice is around 3,202,944 metric tons including post- harvest loss for human consumptions as food, beverages and other industrial usage.

Per capita rice supply per year was 114 Kg according to the latest figures of the Food Balance Sheet with the imported amount of rice. However, the average per capita rice consumption was estimated as 104.5 Kg per year by the Income and Expenditure Survey in Sri Lanka that has been done for the same period.

2.3. Paddy cultivation by administrative district in Sri Lanka

Paddy cultivation by administrative districts in Sri Lanka is given in the following table. The area of the paddy tracks which have been cultivated at least one cultivated season within last five year period is identified as asweddumized paddy field. There were 859514 hectares of asweddumized paddy lands as 1738472 parcels all over the country. The most prominent district for paddy cultivation is Anuradhapura district and it was 15.3 percent of the total paddy land area (see Table 2.1). Kurunegala and Ampara districts were following the Anuradhapura district as second and third districts but the highest number of paddy parcels has been reported in Kurunegala district and it was 343659 parcels. Batticaloa and Polonnaruwa district were also be conspicuous for paddy cultivation and percentages of the land areas were 8.3 and 8.2 of the total asweddumized area respectively. The total land areas of those five districts were also comparatively higher than the other districts of the country (see Figure 2.1 below).

However, the asweddumized area of the paddy lands in western province was not a significant amount and it was around 4 percent of the total paddy land areas of the country.

Table 2.1 : Asweddumized area of paddy cultivation by Administrative districts in Sri Lanka - 2021

Administrative district	No. of Parcels		Asweddumized area		
	No.	%	Acres	Hectares	%
Colombo	17742	1.0	11708	4738	0.6
Gampaha	60807	3.5	35204	14247	1.7
Kalutara	52832	3.0	38830	15714	1.8
Kandy	54495	3.1	35680	14439	1.7
Matale	67829	3.9	54842	22194	2.6
NuweraEliya	28930	1.7	16037	6490	0.8
Galle	50184	2.9	39313	15910	1.9
Matara	69859	4.0	40885	16546	1.9
Hambantota	64163	3.7	100965	40860	4.8
Jaffna	31288	1.8	28165	11398	1.3
Mannar	20998	1.2	52186	21119	2.5
Vavuniya	20646	1.2	51315	20767	2.4
Mullaitivu	20010	1.2	54124	21904	2.5
Kilinochchi	17926	1.0	69001	27924	3.2
Batticaloa	58834	3.4	175677	71096	8.3
Ampara	88928	5.1	214383	86760	10.1
Trincomalee	45905	2.6	100491	40668	4.7
Kurunegala	343659	19.8	216661	87682	10.2
Puttalam	49545	2.8	57266	23175	2.7
Anuradhapura	241896	13.9	324144	131179	15.3
Polonnaruwa	76377	4.4	174802	70741	8.2
Badulla	93386	5.4	76365	30904	3.6
Moneragala	70065	4.0	93729	37932	4.4
Ratnapura	48279	2.8	42904	17363	2.0
Kegalle	43889	2.5	19182	7763	0.9
Total	1738472	100.0	2123859	859514	100.0

2.4. Mode of irrigation for cultivation

Water supply for the cultivation is very essential factor. The water supply should be systematically controled at the each stage of the paddy cultivation from broadcasting to harvesting of the paddy. The paddy cultivation is done under three types of irrigation modes in Sri Lanka namely major, minor and rainfed. The paddy cultivation is totally performed by the rain fall water under the rainfed irrigation mode and it is very famous in wet zone of the country but somewhat difficult to cotrol the water supply to the paddy field by this method. If it is available a water tank (Wewa) or reservoir to supply water with formal controlling system for the paddy field and the capacity of the water tank is sufficient only for less than 200 acres of paddy land area cultivation then it is called minor irrigation mode. Therefore the water tanks and resevoirs those can be supplied water for the cultivation grater than or equals to 200 acres of paddy land is identified as major irrigation mode.

Boundaries of the systematic administrative system of the country are illustrated in the following charts and sampling procedures can be easily explained by using these three charts.

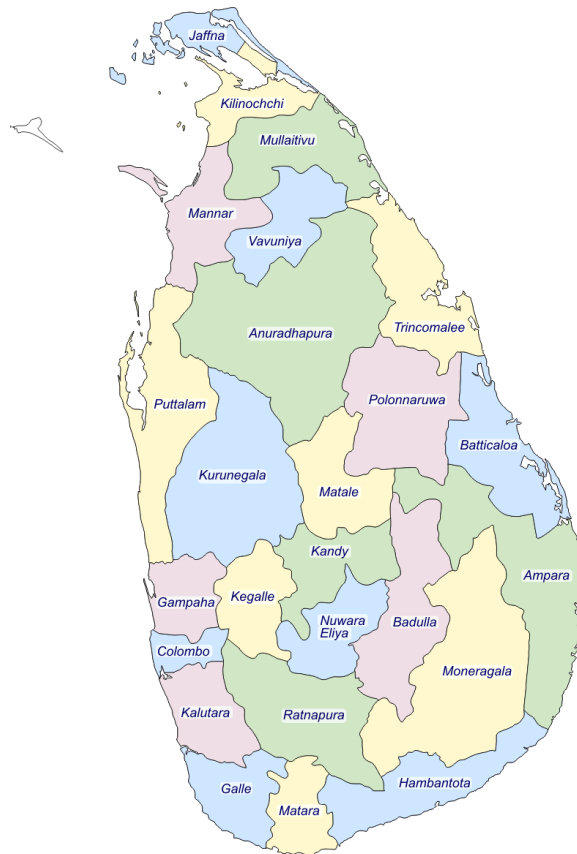


Figure 2.1 : Administrative Districts in Sri Lanka

The Anuradhapura district that is most prominent to the paddy cultivation is taken to explain the cultivation pattern with hypothetically designed irrigated regions. The Anuradhapura district map is given in the following Figure 2.2.



Figure 2.2 : Divisional Secretariats in Anuradhapura Districts in Sri Lanka

Distribution of the irrigation modes of Horowpathana Divisional Secretariat in Anuradhapura District is illustrated in Figure 2.3 below.

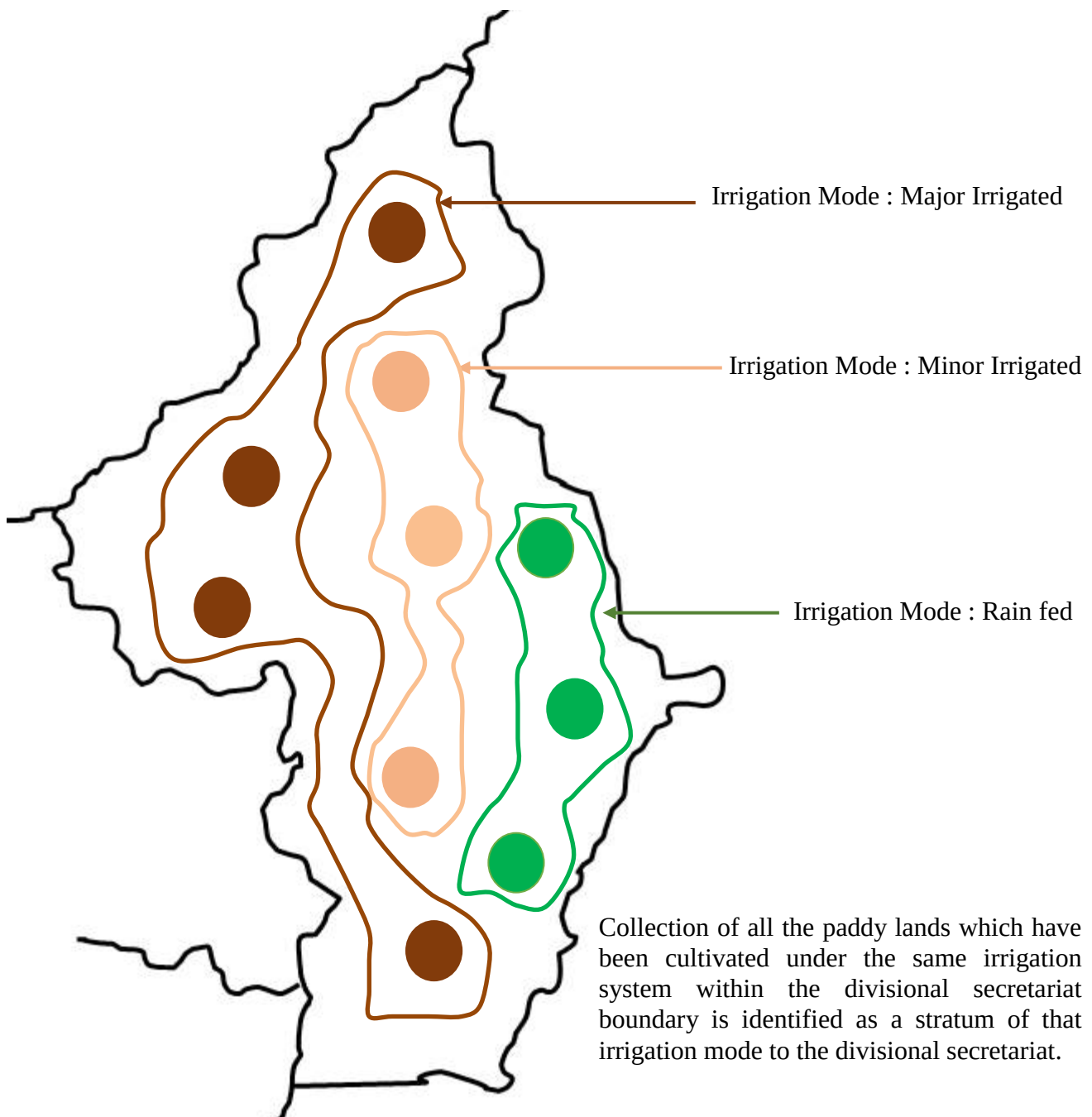


Figure 2.3 : Irrigation modes in Horowpathana Divisional Secretariat in Anuradhapura District

Collection of all the paddy lands which have been cultivated under same irrigation system within the divisional secretariat boundary is identified as a stratum of that irrigation mode.

2.5 Elementary practice of the sampling design

The paddy cultivation is mainly done under three irrigation modes namely major, minor and rainfed in both the first (Maha) and the second (Yala) main agricultural seasons in Sri Lanka as explained above. Each irrigation mode in the divisional secretariat is concerned as a stratum. Therefore the selection of the Divisional Secretariat and the irrigation mode are the first and the second sampling stages to the

overall sampling process. Maximum number of strata is three to the paddy cultivating divisional secretariat and minimum is one for it.

Paddy fields are available and paddy cultivations are done in 324 divisional secretariats although the total number is 335 in country (newly created five DS divisions in Nuwara Eliya district is not yet functioning). However no paddy fiels in following Divisional Secretariats in Sri Lanka.

<u>Administrative District</u>	<u>Divisional Secretariat</u>
Colombo	Colombo
	Sri Jayawaradanapura Kotte
	Thimbirigasyaya
	Dehiwal
	Rathmalana
	Moratuwa
	Negambo
Gampaha	Kelaniya
	Nuwara Eliya (Paddy lands are available but no cultivation)
Nuwara Eliya	
Batticaloa	Kattankudy
Puttalam	Kalpitiya

Althose the paddy lands are available in Nuwar Eliya Divisional Secretariat (DS) Division, paddy cultivation is not done but potetos and other vegetables are cultivated by cultivators to obtain more economic benefits.

2.51 Mathematical way of thinking and notations

D_i is the i^{th} Divisional Secretariat Division and variation of i is from 01 to 320 if all the DS divisions are drowned on the map of Sri Lanka with correct boundary demacations. Numbering system is already activated in each DS division in a desultorily and geographically serpentine manner, as encoded in administrative coding system.

S_j is the j^{th} Stratum of the i^{th} DS division and there are only three maximum irrigation modes, hence j is varring from 1 to 3 in the DS division.

Then $\sum_i \sum_j D_i S_j$ is represented the area sample frame and all the paddy land plots of the $D_i S_j$ are the sample units to the j^{th} stratum of j^{th} DS.

The divisional secretariat division is further divided in Grama Niladari (GN) divisions with clear demarcation boundaries for the administrative purposes. Each aswedumized paddy land plot in the GN

division has been taken to the registry by means of it's active owner or operator or cultivator with their names and that registry is called P₁ register. The P₁ register is the village level sub frame of the area sample frame and collection of each and every P₁ registers are finally concerned as area sample frame. The important identification information and other descriptive statistics such as name of the operator, name of the paddy tract, irrigation mode of cultivation, acreage of the paddy land plot have also included under the P₁ register.

P_{1 (ih)} is the register for the h^{th} GN division of the i^{th} DS division and $P_{1 (ih)} = \sum_j \sum_k L_{ihjk}$

$j = 1, 2, 3$ for Major, Minor and Rainfed $k = 1, 2, \dots$ number of plots in the GN division
 $h = 1, 2, \dots$ number of GN divisions in the i^{th} DS division.

L_{ihjk} is the k^{th} land plot in the j^{th} stratum in the h^{th} GN division in i^{th} DS division. It is symbolized the sample unit.

The P₂ register is prepared to the DS division and it is included the summary information of P₁ register. The P_{2 (i)} register has seperated into three parts for major, minor and rainfed irrigation modes and one line has been allocated for each irrigation mode in each GN division if it is availabe asweddumized paddy land plots in that irrigation mode.

P₂ register for the i^{th} DS division $= \sum_h \sum_j \sum_k L_{ihjk}$ and it can be concerned as DS based sub sample frame.

Now it can be prepared summary statement for the area sample frame to the paddy crop cutting survey as follows.

$$\sum_i \sum_j D_i S_j \approx \sum_i \sum_h \sum_j \sum_k L_{ihjk}$$

is the complete sample frame to the paddy crop cutting survey in Sri lanka and collection of the list of paddy fields in aech GN division. These lists are prepared to the five year period and updated twice a cultivated season to ascertain the actual sown area and the harvested area.

3. Irrigation for cultivation

3.1 Pattern of the monsoon winds and rains in Sri Lanka

Monsoon rains, convectional rains and cyclonic rains are the three main sources of rains during the year in Sri Lanka. It has a typical time period for monsoon rain and conventional rain but cyclonic rains can be expected any time through out the year. Normally, conventional rains are activated in transitional periods of two different monsoon winds as north eastern monsoon winds to south western monsoon winds and vice versa. (Refer “Effects of the Atmospheric Conditions and Ecoclimate to the Agriculture in Sri Lanka” for further knowledge)

The south-western monsoon brings rain to the south-west of Sri Lanka (Wet zone) between May and September prominently, while the dry season in this region runs from October to March. In the north and east coastal regions (Dry zone), the monsoon brings wind and rain from October to March under the influence of north eastern monsoon winds. In the mean time, the intermediate zone of the country depends on above two types of monsoon winds and rains with different climatic pattern.

Table 3.1 : Cultivation seasons with monsoon winds by months of calendar year

	Calendar year	Period by calendar months	Type of monsoon rain	Monsoon rain activated region	Season of cultivation	
					main season	with intermediate seasons
Cultivated season	This year	October to March	North eastern monsoon	Dry zone	Maha	First season
						Fourth season
	Next year	April to September	South western monsoon	Wet zone	Yala	Second season
						Third season

3.2 Agro ecological regions in Sri Lanka

Sri Lanka has a heterogeneous agro-ecological environment and therefore the country has traditionally been classified into three climatic zones as Wet Zone, Dry Zone and Intermediate Zone on rainfall distribution. The wet zone covers the south-western region including the central hill country and receives relatively higher annual average rainfall over 2,500 mm without pronounced dry periods. The dry zone covers predominantly the northern and eastern part of the country, being separated from the wet zone by the intermediate zone. The annual average rainfall of the dry zone is less than 1,750 mm and the dry season is influenced from May to September. The Intermediate zone receives a mean annual rainfall between 1,750 to 2,500 mm with a short and less prominent dry season. See the following Figure 3.1 for the geographical illustrations of the regions according to the administrative districts.

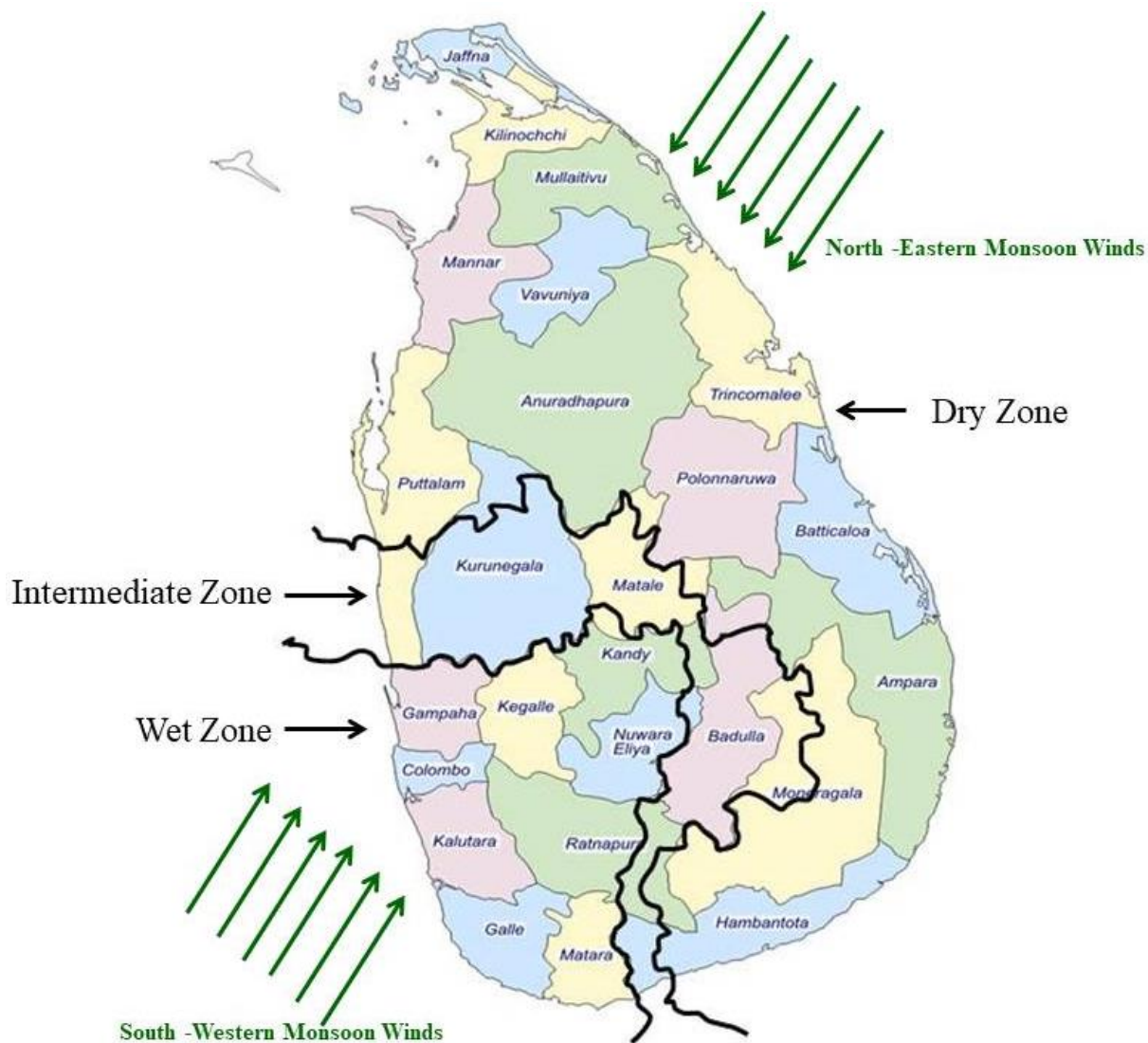


Figure 3.1 : Agro-Ecological Regions in Sri Lanka

3.3 Identification of climatic zones by divisional secretariats

The country has divided into 25 administrative districts and those administrative districts have further divided as divisional secretariat (DS) divisions for the administrative purposes. There are 335 divisional secretariate divisions in Sri Lanka.

Colombo, Gampaha, Kalutara, Galle, Matara and Kegalle districts are totally belonged to the wet zone and Jaffna, Mannar, Vavuniya, Mullaitivu, Kilinochchi, Batticaloa, Ampara, Trincomalee, Anuradhapura and Polonnaruwa districts are absolutely belonged to the dry zone. Almost all the areas of the Kurunegala district can be concerned as intermediate zone but Giribawa divisional secretariat division is belonged to the dry zone. The pattern is same to the Badulla district. All the areas except Mahiyanganaya divisional secretariat division is belonged to the intermediate zone. The summary information is given in the following Table 3.2.

Table 3.2 : Number of DS divisions to the climatic zones by administrative district

Administrative district	No. of DS Divisions by Climatic zone			
	Wet	Intermediate	Dry	Total
Colombo	13			13
Gampaha	13			13
Kalutara	14			14
Kandy	19	1		20
Matale	1	8	2	11
NuweraEliya	4	1		5
Galle	22			22
Matara	16			16
Hambantota		6	6	12
Jaffna			15	15
Mannar			5	5
Vavuniya			4	4
Mullaitivu			6	6
Kilinochchi			4	4
Batticaloa			14	14
Ampara			20	20
Trincomalee			11	11
Kurunegala		29	1	30
Puttalam		5	11	16
Anuradhapura			22	22
Polonnaruwa			7	7
Badulla		14	1	15
Moneragala		5	6	11
Ratnapura	13	4	1	18
Kegalle	11			11
Total	126	73	136	335

4. Sample selection to the paddy crop cutting survey

The average yield as well as the crop grown pattern and the quality of the plants can be varied with the irrigation mode. Heterogeneous catalysis is comparatively very higher for the cultivations of the rainfed irrigation because rainfall cannot be controled and therefore it is very difficult to manage rain water at the paddy cultivation. Some times, adequate rain water has not been received to the essential period of time to the crop growings. So, it should be given more weight to the rainfed irrigation mode when it is arranged sample allocations.

4.1 Sample selection

Total sample size is 4000 paddy plots to the cultivated season of the paddy crop cutting survey in Sri Lanka. 4000 paddy fields are selected by using the P2 summary reports of the DS divisions at the primary stage under the system of proportional allocations. A sample of P1 registers are selected at this first stage. Further, the sample units are randomly selected by the names of the cultivators as represent each rice paddy cultivated DS division by using the P1 register.

Stage I : Bonus allocation

The primary objective of the sample section is to estimate the paddy production to the Divisional Secretariat division level and therefore bonus samples are allocated to the each and every paddy cultivated DS division at the cultivated season accoding to the irrigation mode as following table to maintain the adequate sample size to the division.

Table 4.1 : Bonus experiments to the DS division by climatic zone

Climatic zone	Irrigation modes in Maha Season			Irrigation modes in Yala Season		
	Major	Minor	Rainfed	Major	Minor	Rainfed
Wet	2	2	4	2	2	3
Intermediate	2	2	2	2	2	2
Dry	2	2	3	2	2	2

Stage II : Proportional allocation

The proportional sample allocation system is used to select the remaining sample units to the DS division (Total sample allocatio – Bonus allocations) by concerning the sown extent under each irrigation mode. The variation of the yield is comparatively higher in rainfed irrigation mode due to uncontrollable system of water supply and therefore priority is given to this system when is practising the proportional allocation system under both stage I and stage II.

The percentage of proportional allocation is 40 to the rainfed irrigation mode as well as major irrigation mode for Yala season and the remaining 20 percent of the sample is allocated to the minor irrigation mode in the same monsoon season as following table.

Table 4.2 : Percentage distribution of proportional allocations to the DS division by climatic zones and monsoonseason

Irrigation mode	Monsoon Season (%)	
	Maha	Yala
Major	30	40
Minor	40	20
Rainfed	30	40

Stage III : Selection of sample paddy lands to the DS division

It has been already prepared the village level summary information of paddy land extent under each GN division in the DS division (P2 register) with formal numbering system according to the irrigation mode separately. Likewise, the sample sizes of each DS division can be decided at the end of the above stage II in each irrigation mode. Then the villages or paddy tracks (Kumburu Yaya) can be selected under simple random sampling technique by the names of formal numbering system of the P2 register.

After selecting the villages or paddy tracks, it can be found the selected GN divisions and the list of paddy lands with the names of the cultivators are available for each GN division (P1 Register). Secondly, the sample paddy lands are also selected under simple random sampling techniques by using the P1 register. P1 registers are available in districts offices and therefore this step can be completed to the staff of the district office.

4.2 Selection of the experimental paddy plots

The list of sample paddy lands is given to the enumerator who is the responsible officer for this very important and noteworthy task. He goes to the relevant paddy field and meet the cultivator with the sufficient time gap to the harvesting date. The approximate date of the harvesting is taken from the cultivator and secondly visit to the paddy field again two weeks before that approximate harvesting date.

The rough sketch of the selected paddy field to the sample is drawn in his field note book with the number of discernible square areas (Liyali) and demarcation boundaries by the enumerator. Then study the water supply and drainage pattern of the paddy field and numbering the discernible square areas to the ascending order from lowest level to highest level on serpentine manner as shown in the following Figure 4.1.

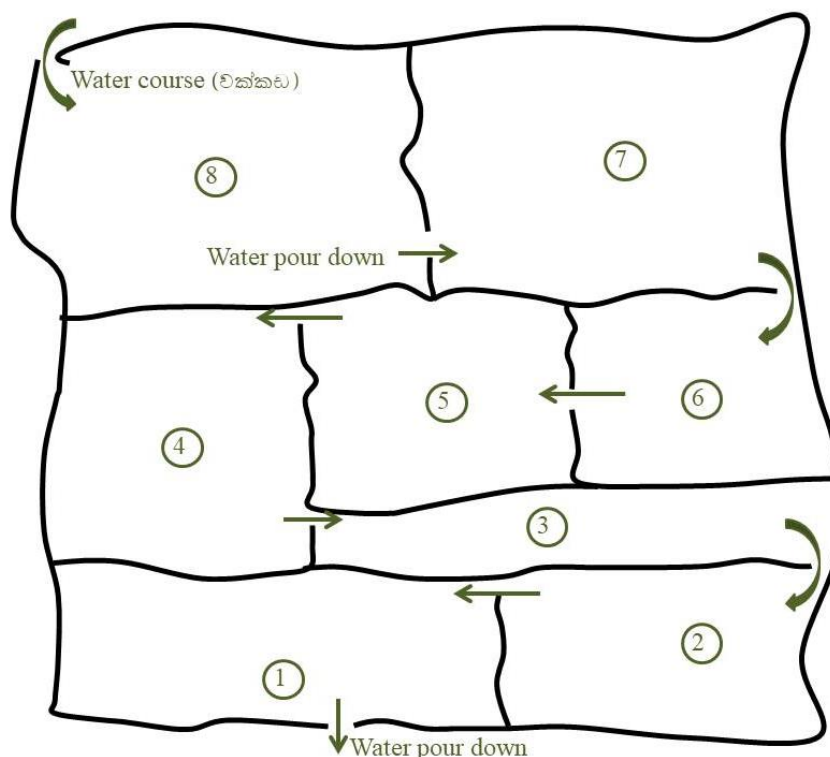


Figure 4.1 : Sketch lay out of the selected paddy field

A suitable discernible square area which is relevant to the above numbering system is randomly selected by using the simple random sampling techniques. The “Book of random numbers for the seloection of sample units in crop cutting survey” is available with enumerators to do this.

Then, the gross length and the gross width of the randomly selected discernible square area are measured by using the meter tape and recorded it in the field note book. Secondly reduced the amount of 503 cm (5 meters and 3 centimeters or 17 feet) from both gross lenth and the gross width and restablished the net length and net width as integers by truncating decimal values of the meter units.

Measure	Gross value	Net value	Random number
Length	L_g	$L_n = L_g - 5.03$	$R_l ; (\leq L_n)$
Width	W_g	$W_n = W_g - 5.03$	$R_w ; (\leq W_n)$

The first random number (R_l) which is relevant to the net lenth (L_n) has to be selected by using a computer software or a random number table and the second random number (R_w) for the net width (W_n) is also been selected according to the same procedure. Both numbers should be less than or equals to thenet values. Then, the first point of the sample plot is marked in the south west corner of the discernible square area as depicted in the following figure. It should be measured the distance of R_l along the long ridge and then R_w distance shold be gone parallel to the short ridge to obtain above first point. Both measurements are taken to the forward directions from the south west corner. This is the first point of the sample plot.

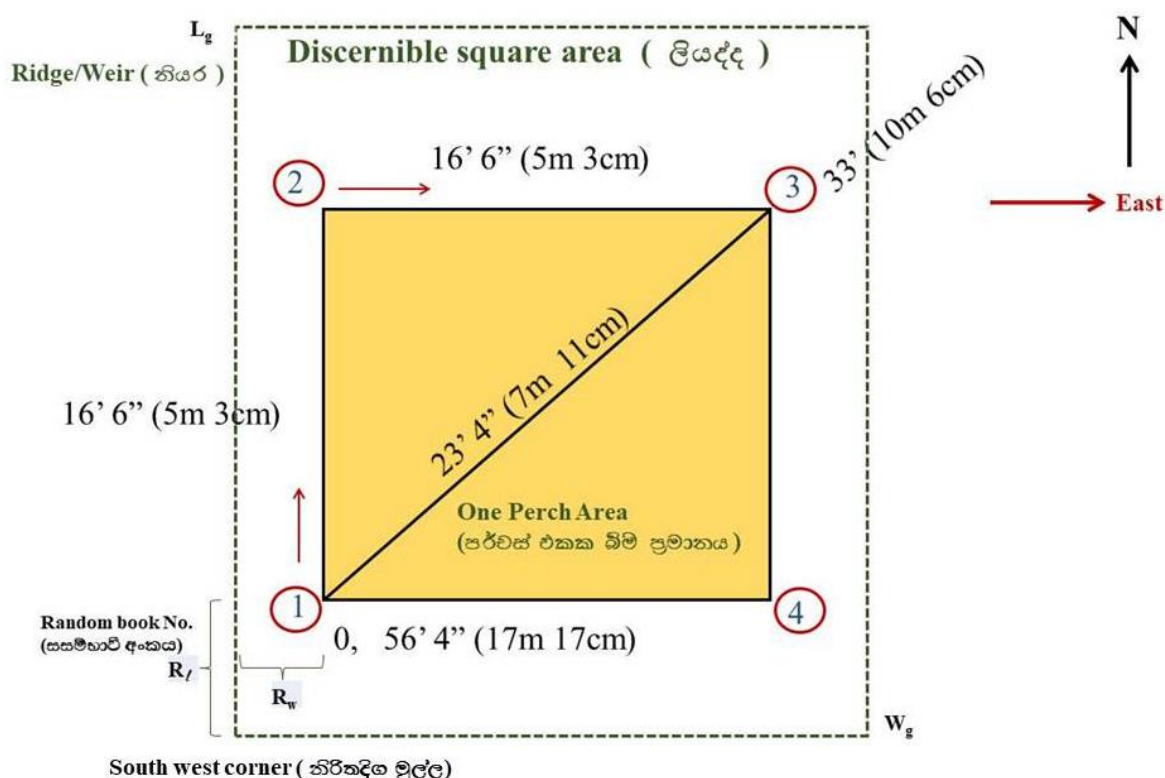


Figure 4.2 : Sketch lay out of the selected discernible square area

The first stick is fixed in this point and measure 5.03 m (16 feet and 6 inches) distance parallel to the forward direction of the long ridge. This is the second location and then turn to the right by right angle with lengthening the metre tape as illustration in the above picture. The third location is temporary selected as the length from first point to the second location and second location to the third location. Then turn back to the first point along the diagonal line with furthermore lengthening the metre tape. The last reading of the metre tape should be 17m and 17cm when it is reach to the first point again along the diagonal line and the third location can be adjusted to fulfil it. The first point, second location and the third location are represented a right angled triangle. Finally, the intermediate place of the metre tape which is hold at the second location should be thrown away over the diagonal line to the opposite direction for the demarcation of the fourth location.

Four straight sticks have to be fixed on these four locations and these sticks should be connected by a cord over the paddy plants to demarcate the square area of one perch extent. It is a quadrangle and the yield of this area should be harvested.

However, the plot size is decided by concerning the extent of the discernible square area and if it is not suitable to select above standerd plot due to insufficient length and width of the discernible square area then the experimental plot can be selected as the measurements given in the following Table 4.3.

Table 4.3 : Experimental parcel sizes of paddy crop cutting survey

	Length		Width		Diagonal		Perimeter of the triangle		Area	
	(IS units)	(metric units)	(IS units)	(metric units)	(IS units)	(metric units)	(IS units)	(metric units)	Sq feet	Perch
1	16' 6"	5.03 m	16' 6"	5.03 m	23' 4"	7.11 m	56' 4"	17.17 m	272.25	1
2	24' 9"	7.54 m	11' 0"	3.35 m	27' 1"	8.25 m	62' 10"	19.15 m	272.25	1
3	16' 6"	5.03 m	8' 3"	2.51 m	18' 5"	5.62 m	43' 2"	13.17 m	136.125	0.5
4	12' 4.5"	3.77 m	5' 6"	1.68 m	13' 6.5"	4.13 m	31' 5"	9.58 m	68.0625	0.25
5	4' 0"	1.22 m	3' 0"	0.91 m	5' 0"	1.52 m	12' 0"	3.66 m	12	0.044

4.3 Harvesting of the experimental paddy plots

All the rice plants can be harvested manually by using the sickle and it is the most suitable way for safely collection of all the paddy seeds by minimising the wastage as exhibit in the below Figure 4.3 below.

The paddy pods which were accidentally left on the harvested plot by the reaper due to unawares avoidance should be picked up by another person to ascertain correct harvest of the sample plot.



Figure 4.3 : Reaping the paddy plants manually

The harvested rice plants of the crop cutting experiment have to be collected to the gunny bag and wrapped gunny bag with rice plants should be brought to the threshing floor. The farmer believes that this activity is a very adorable work. Therefore it is a responsibility of the relevant officer to do this very carefully.

A scaffold is constructed as illustrated in the following Figure 4.4 to trample the harvest manually.



Figure 4.4 : Threshed the paddy manually by using the scaffold

Paddy can be threshed by friction involves trampling or stomping on a pile of harvested paddy to separate the grains from the straw at the threshing floor as above illustration



Figure 4.5 : Removing husks, chaff and other impurities of the paddy

Winnowing fan (Kulla) and other various methods can be implemented to separate paddy grains by removing husks(empty grains), chaff and other impurities. The harvested paddy yield should be opened to the sun shine for the improvement of aridity.

4.4 Measuring the harvested yields

The harvested paddy yield of the experimental plot has to be measured by using the sheer set after cleaning and dried adequately.



Figure 4.6 : The harvested paddy yield is ready to measure

The sheer is belonged to the standard imperial system of measurements and the inner volume of the standard sheer is 1136.6 ml. The second piece is exactly half of the first piece and recurring for each as given in following Table 4.4. It has included six parts to the sheer set and the volume of the smallest piece is 35.517 ml. However, it is one over thirty two ($\frac{1}{32}$) of the standard sheer. Further, 32 sheers are called one bushal (36.368 litres) and it is the preparative measurement for the paddy production.

Table 4.4 : Sizes of the sheer set by metric units

System of Imperial Units	Metric Units
1 Sheer = Quart (සේරුව/ක්වාට්/තැලිය)	1.1366 <i>l</i>
$\frac{1}{2}$ Sheer = Pint (පයින්ට්)	568.261 <i>ml</i>
$\frac{1}{4}$ Sheer = Quarter (ක්වාර්ටර්)	284.130 <i>ml</i>
$\frac{1}{8}$ Sheer = Gill (ගිල්)	142.065 <i>ml</i>
$\frac{1}{16}$ Sheer = 2.5 Liquid Ounce (20 Drams)	71.033 <i>ml</i>
$\frac{1}{32}$ Sheer = 10 Drams	35.517 <i>ml</i>
1 Gallon = 4 Sheers (කුරුණිය/ලාස්)	4.546 <i>l</i>
1 Bushal = 32 Sheers (බුසල)	36.368 <i>l</i>
1 Bushal = 1.24 Cubic Feet (සත අඩ්)	

4.5 Record the result of the experiment with the other related information

Identification information of the experimental land plot is essential when it is record the result of the paddy crop cutting experiment to identify the geographycal location and the name of the cultivator. Those information is collected to the special form (CC3) with cultivation related information. The seed variety, usage of fertilizer, usage of insecticides and pesticides, usage of weedcides as well as irrigation mode is very important information to estimate the paddy production. In addition, land extent, tenureship of the paddy land, expected yield are also reported to the above form.

5. Estimation of the paddy production

The most important information to the calculation of paddy production is sown area in addition to the experimental results. The cultivator gives it and he decided it by concerning the legal documents of the paddy land as well as amount of the seed paddy used to the cultivation. This sown area which is disclosed by the farmer is taken as the gross sown area but it can be varied due to various reasons. The sown area can be differ with the number of discernible square areas and the width of the weir and those things can be varied according to the irrigation mode as well as agro ecological regions in the country. Therefore area correction factors have been calculated based on administrative district to solve this problem and this adjusted sown area is called net sown area.

Further, measurements of the yield in experimental plot area is also very important factor to the accurate estimation. Volume expansions of the paddy seeds are comparatively lower than the weight increase when it is moisten the paddy seeds. Therefore the volume is measured by using sheer set instead of measuring the weight and later on convert the production in to the metric tons. The basic units of the measurement to the paddy production is bushels in imperial units. Averagely, it has 20.8 Kgs per bushel for dried paddy but little variation can be expected according to the seed variety.

The average yield is estimated for each irrigation mode separately in the Divisional Secretariat division and then the net production is estimated by using the district area correction factor to the same locality. However, the paddy productions are published to the administrative district to minimise the sampling errors.

5.1 Estimation of the annual paddy production and rice requirements of consumers

The paddy production of the calendar year 2021 is the total production of 2020/21 cultivation year and 770240 hectares has been cultivated at the Maha season of that cultivation year. It is around 90% of the asweddumized area (859,514 hectares). However, the cultivation of paddy lands to the Yala season is significantly lower at the Yala season (58%) according to the following Table 5.1. Although the harvesting loss is minimum when it is harvested manually, it is significant amount if the combine harvester is used for harvesting. But it cannot be found proper method to measure harvesting losses and assumption is 6 percent of the total production. The net value of the paddy production in the year 2021 was 4,709,753 metric tons and it is around 3,202,632 metric tons of raw rice. The usage of rice flour for other food productions are very famous in the country and significant amount of raw rice has also been used as raw material in beverages industry too. What so ever, no formal information to prove the amount of usage.

However, the consumption demand of the raw rice in the year 2021 was 2,334,384 metric tons in Sri Lanka but raw rice production in the same year is 2,702,632 metric tons according to the survey estimations. It is better to prosecute the following table for detail information.

Table 5.1 : Paddy production and consumption pattern in Sri Lanka - 2021

Category	2020/21 Maha	2021 Yala	Year 2021
Sown Extent (Hectares)	770,240	501,792	1,272,032
Paddy Production (MT)	3,061,394	2,088,486	5,149,880
Harvesting loss (6%)	183,684	125,309	308,993
Net Harvesting amount	2,877,710	1,963,177	4,840,887
Seed requirements to the season	79,404	51,730	131,134
Net Production of Paddy	2,798,306	1,911,447	4,709,753
Raw Rice Production	1,902,848	1,299,784	3,202,632
Raw Rice for other food processing (20%)	380,570	259,957	640,526
Raw Rice Importations (annual)	147,091		147,091
Raw Rice Exportations (annual)	6836		6836
Total Rice Supply	1,662,534	1,039,827	2,702,361
Raw Rice for food	1,662,534	1,039,827	2,702,361
Projected population at the end of 2021	22360000	22360000	22360000
Per capita rice supply per Season (Kg)	74.35	46.50	120.86
Rice Consumption per person per month Kg)	8.7		
Rice Consumption per person per Season	52.2		104.4
Consumption demand of the rice to the population (MT)			2334384
Locally supplied for consumption purposes (MT)			2,562,106

Population in Sri Lanka is expected to reach **22.36 Million** at the end of 2021, according to Trading Economics Global Macro Models and analysts expectations though it was estimated as 22.16 million by DCS.

Further, locally produced raw rice is used for the animal feeds on a large scale in the country but it is very difficult to measure the amount because the purpose of purchasing is not categorized by the intermediate dealers.

5.2 Future planning and solutions to the crisis

Although the identified areas as paddy lands in the country are considerable amount, a significant proportion of them have not been cultivated within long period of time due to various reasons. The paddy lands which were never cultivated in previous ten consecutive cultivated seasons are identified as abandon paddy lands and those land areas have not been included to the P₁ Register. If it is cultivated some time or other in previous five year period then the land is concerned as asweddumized paddy land.

The asweddumized areas of paddy lands are given in the following table by administrative districts in descending order with the average yields for both seasons in 2021.

Table 5.2 : Asweddumized area of paddy cultivation and expected production by administrative district in Sri Lanka – 2021

Administrative district	Asweddumized extent		Average yield		Paddy production from (80% Maha and 50% Yala) of the Asweddumized Area						
			(Kg per Hactare)		2020/21 Maha season (80%)			2021 Yaha season (50%)			2021
	2020/21	2021	Paddy (MT)	Cumulative value		Paddy (MT)	Cumulative value		Rice production (Cumulative value)		
	Maha	Yala		Paddy	Rice		Paddy	Rice			
Anuradhapura	324144	131179	5098	4691	535002	535002	363802	307681	307681	209223	573025
Kurunegala	216661	87682	3775	4024	264798	799800	543864	176415	484097	329186	873050
Ampara	214383	86760	4780	5229	331769	1131569	769467	226833	710930	483432	1252899
Batticaloa	175677	71096	3809	4665	216642	1348211	916784	165830	876760	596197	1512981
Polonnaruwa	174802	70741	5220	5441	295415	1643627	1117666	192451	1069211	727064	1844730
Hambantota	100965	40860	5883	5814	192303	1835930	1248432	118780	1187991	807834	2056267
Trincomalee	100491	40668	4250	5267	138272	1974202	1342457	107100	1295091	880662	2223119
Moneragala	93729	37932	4599	5034	139558	2113760	1437357	95474	1390565	945584	2382941
Badulla	76365	30904	4859	5088	120132	2233892	1519046	78621	1469186	999046	2518093
Kilinochchi	69001	27924	3500	4620	78188	2312080	1572214	64505	1533691	1042910	2615124
Puttalam	57266	23175	3746	3950	69452	2381531	1619441	45771	1579462	1074034	2693476
Matale	54842	22194	5669	4639	100655	2482187	1687887	51480	1630942	1109040	2796927
Mullaitivu	54124	21904	4266	5287	74753	2556940	1738719	57902	1688844	1148414	2887133
Mannar	52186	21119	5228	4931	88330	2645269	1798783	52070	1740914	1183821	2982605
Vavuniya	51315	20767	4253	5638	70657	2715927	1846830	58542	1799456	1223630	3070460
Ratnapura	42904	17363	4465	4119	62021	2777947	1889004	35759	1835215	1247946	3136950
Matara	40885	16546	3585	3686	47454	2825401	1921273	30494	1865709	1268682	3189955
Galle	39313	15910	3403	2897	43313	2868714	1950725	23045	1888754	1284353	3235078
Kalutara	38830	15714	3358	3048	42215	2910929	1979431	23949	1912703	1300638	3280069
Kandy	35680	14439	4720	2945	54524	2965452	2016507	21262	1933965	1315096	3331604
Gampaha	35204	14247	3751	2991	42752	3008204	2045579	21306	1955271	1329584	3375163
Jaffna	28165	11398	2549	0	23243	3031447	2061384	0	1955271	1329584	3390969
Kegalle	19182	7763	4163	2991	25853	3057301	2078965	11609	1966881	1337479	3416443
NuweraEliya	16037	6490	5020	3917	26064	3083365	2096688	12711	1979591	1346122	3442810
Colombo	11708	4738	3723	2455	14112	3097477	2106284	5816	1985407	1350077	3456361
Total	2123859	859514	4307	4307	3097477			1985407			

The most prominent district for paddy cultivation is Anuradhapura district as already explain under section 2.3 above. However, paddy cultivation in Maha season is significantly higher than the Yala season in Sri Lanka and no paddy cultivation to the Yala season in Jaffna district. It is not necessary to cultivate total asweddumized areas in each season for the fulfilment of the consumption requirements of the country. All the calculations in the above Table 5.2 have been done under the assumptions of 80 percent of the area is cultivated in Maha season and it is 50 percent for the Yala season. The paddy production of first eight cultivation area dominated districts is sufficient for the annual consumption demands in the country according to the indicators shown in the above two tables.

The vast majority of the areas in paddy cultivation prominent districts are belonged to the dry zone. Therefore it is very important to maintain the formal irrigation system to provide water for the cultivation. An adequate number of irrigated tanks are already available in the areas and all the tanks can be filled by monsoon rains very easily. But the main problem is the gradual reduction of the water

capacity of the tanks due to silting in addition to the vaporisation. Water vaporisation is being gradually increased as a adverse effect of the ceaseless global warming.

The average yield is highest in Hambanthota district for both seasons and the main reason is that the seed varieties (specially AT 362) which have been developed at the Ambalanthota Paddy Research Institute have been used to the cultivations.

Table 5.3 : Asweddumized area of paddy cultivation and expected production by administrative district in wet zone in Sri Lanka – 2021

Administrative district	Asweddumized extent (Hectares)	2020/21 Cultivation year							
		Sown extent (Hectares)		Average yield (Kg per Hactare)		Estimated production (MT)			
		Maha	Yala	Maha	Yala	Paddy			Raw rice
						Maha	Yala	Total	
Ratnapura	17363	13358	10773	4465	4119	77526	71518	149044	101350
Matara	16546	14114	14152	3585	3686	59317	60988	120305	81808
Galle	15910	13511	8548	3403	2897	54141	46091	100231	68157
Kalutara	15714	13019	9766	3358	3048	52769	47897	100666	68453
Kandy	14439	12186	9278	4720	2945	68154	42524	110679	75262
Gampaha	14247	12450	9954	3751	2991	53440	42612	96052	65316
Kegalle	7763	6751	5477	4163	2991	32317	23219	55535	37764
NuweraEliya	6490	4877	2573	5020	3917	32580	25422	58002	39441
Colombo	4738	3857	2776	3723	2455	17640	11632	29272	19905
Total	113210	94123	73297	4021	3228	447884	371903	819788	557456

The pattern of paddy cultivation in wet zone is shown in the above Table 5.3 and it is completely carried on by using rain water in these areas. But a significant proportion of the paddy lands are abandon without cultivation within long period and it may be more than the asweddumized areas of 113210 hectares. It should be taken necessary actions to cultivate these arable lands immediately to solve the food problem in the country with long term plan. Inundation is the main problem in this areas and it is more acute to the Yala season. Cultivated paddy fields were damaged every now and then due to floods and it is pave the way for removing the occupied communities from the paddy cultivations. Therefore it is necessary to introduce special flood controlling system in these areas.

In addition, the younger generation have to be motivated for this national task by given appreciable incentives. Introduction of modern techniques and equipments to this field are very important for the motivations.

Appendix

ජනලේඛන හා සංඛ්‍යාලේඛන දෙපාර්තමේන්තුව
DEPARTMENT OF CENSUS AND STATISTICS

වී අස්වැන්න ඇස්තමේන්තු කිරීමේ සම්කෘතිය }
Crop Estimating Survey on Paddy }

මහ/ Maha

යල/ Yala

දිස්ත්‍රික්කය/මහවැලි කලාපය District / Mahaweli System ගො. මේ. මධ්‍යස්ථානය A.S. Centre ග්‍රා. නි. කො. අංකය/ G.N. Division's No.	ප්‍රා. මේ./ප්‍රා. උ. දී./කළමනාකරණ කොට්ඨාසය D.S./A.G.A. / Block Manager's Division ග්‍රා. නි./ ඒකක කළමනාකරණ කොට්ඨාසය G.N./ Unit Manager's Division ගම/ යාය/ කන්දම Village/ Tract/ Kandam
භාරිකාරික ක්‍රමය/ Mode of Irrigation -	වැඩුල්ල/ Major කුඩා/ Minor වර්ෂා මෝසම/ Rainfed
කපන නිලධාරියාගේ නම/ Crop Cutting Officer's Name -	
ලිපිනය/ Address -	

	I කැබැල්ල/ Parcel I	II කැබැල්ල/ Parcel II
1. කුසුරු කැබැල්ලේ නම/අංකය Name of the Parcel/ No		
2. ගොඩියාගේ නම Name of the Farmer		
3. කුසුරු කැබැල්ලට ළඟා වන අන්දම Mode of approach to the parcel		
4. කුසුරු බැලීමට ගිය පළමු දිනය Date of first visit to the field		
5. ගොඩම කැපීමට නියමිත දිනය හා වේලාව Date and time fixed for harvesting		

6. කැබැල්ලේ වැසුරු ප්‍රමාණය Extent sown in Parcel		I කැබැල්ල/ Parcel I			II කැබැල්ල/ Parcel II		
		අ. A.	රු. R.	පරි. P.	අ. A.	රු. R.	පරි. P.
7. කැබැල්ලේ ඇඟි ලිය දි ගණන No. of Liyaddas in the Parcel							
8. ලිය දී තේරීම/ Selection of Liyaddas							
සසම්භාවී අංක පොතේ/ in Random Book		රේඛාව/ Row					
		තීරුව/ Column					
		අංකය/ Number					
9. ලිය දිගේ දිග අඩ/මීටර් Length of Liyadda ft./ metres							
10. ලිය දිගේ පළල අඩ/මීටර් Breadth of Liyadda ft./ metres							
11. අඩ 17/ මීටර් 6 අඩු කර දිග* Length Minus 17 ft./6 metres*							
12. අඩ 17/ මීටර් 6 අඩු කර පළල* Breadth Minus 17 ft./6 metres*							
13. ප්‍රමාණය තේරීම Selection of Plot							
සසම්භාවී අංක පොතේ/ in Random Book		රේඛාව/ Row					
		තීරුව/ Column					
		අංකය/ Number					
		දිග Length	පළල Breadth		දිග Length	පළල Breadth	
14. සංඛ්‍යා කාරක සසම්භාවී පොතේ අංකය / Random Book Number							

දිනය/ Date:-

අත්සන/ Signature

ගොඩියා නමුත් විය පළමු දිනට මෙම ආකෘතිය පුරවා දිස්ත්‍රික් ලේකම් වෙත යවන්න. / Fill this form on the day of the first visit to the field and post it immediately to the District Secretary

* උඩරට බුදුම ගඟේ ලිය දිග 16 1/2" x 16 1/2" ට වඩා කුඩා ප්‍රමාණයේ පොතේ ඇති ලිය දිග හා පළල අඩු කර ගනු ලැබේ. / Where smaller experimental plot sizes are used in terraced fields, the actual length and breadth of these plots should be deducted.

ජනගණන හා සංඛ්‍යාලේඛන දෙපාර්තමේන්තුව
DEPARTMENT OF CENSUS AND STATISTICS

ජී. සී. 1 ආකෘතිය පුරවා රේඛණේ සති 30 පසුව මෙම ආකෘතිය පුරවා දිය යුතුය.
වෙලාව රේඛණේ යුතුය.

This Form should be sent to G. A. two weeks after the despatch of the Form C. C. 1.

රා. සේ. පි.

සංඛ්‍යා ලේඛන අංශයේ අවධානය සඳහා

දියාපති.

ආකෘතිය සී. සී. 1
Form C. C. 2

වී අස්වැන්න ඇස්තමේන්තු කිරීමේ සමීක්ෂණය
CROP ESTIMATING SURVEY ON PADDY

වසර Maha 20 20.....

යල/ Yala 20.....

ප්‍රා.ලේ/ප්‍රා. හා. හි. කොට්ඨාශය
D.S./ A.G.A. DIVISION

ග්‍රා.
Village

	I පැර්සල් Parcel I	II පැර්සල් Parcel II
1. කුඹුරු පැර්සල් නම / Name of Parcel		
2. වගාවකරු නම / Name of Cultivator		
3. සී. සී. 1 ආකෘතියෙහි දැක්වූ කපන දිනය හා වේලාව /Date and time of harvesting reported on Form C. C. 1.		
4. ප-සංස්කරණ දිනය හා වේලාව / Revised date and time	දිනය / Date වේලාව / Time	

කපන නිලධාරියාගේ / Crop Cutting Officer's —

නම / Name :

තනතුර / Designation :

දිනය / Date :

අත්සන / Signatures.

H 944603 - 15,000 (2015/16) ශ්‍රී ලංකා ජනගණන දෙපාර්තමේන්තුව

DEPARTMENT OF CENSUS AND STATISTICS

වි අස්වැන්න ඇස්තමේන්තු කිරීමේ සමීක්ෂණය } මහ Maha 20...../20.....
Crop Estimating Survey on Paddy } යල Yala 20.....

ආකෘතිය සිසි
 Form CC } 3
 (F2*S. & E.) 09/81

අනුක්‍රමික අංකය }

--	--	--	--	--

(හිස්ව තබන්න / Leave Blank)

I කොටස / PART I							
A1. පරිපාලන දිස්ත්‍රික්කය Administrative District } A2. ප්‍රාදේශීය ලේකම් කොට්ඨාසය Divisional Secretariat Division } A3. ගොඳේ මධ්‍යස්ථානය A.S. Centre } A4. මහවැලි කලාපයක් නම්, If Mahaweli Zone, මහවැලි කලාපය Mahaweli Zone } කළමනාකරණ කොට්ඨාසය Block Manager's Division } ඒකකය Unit Number } A5. ග්‍රාමී. කොට්ඨාස අංකය හා නම G.N. Division Number & Name } A6. ගම/ යාය/ කන්දම Village/ Tract/ Kandam } A7. වාරිමාර්ග ක්‍රමය Mode of Irrigation (අදාළ කේතය කට කරන්න/ Encircle the relevant code) <table border="1" style="margin-left: 100px; border-collapse: collapse; text-align: center;"> <tr> <td style="padding: 5px;">විශාල Major</td> <td style="padding: 5px;">සුළු Minor</td> <td style="padding: 5px;">වර්ෂා වේෂිත Rainfed</td> </tr> <tr> <td style="padding: 5px;">1</td> <td style="padding: 5px;">2</td> <td style="padding: 5px;">3</td> </tr> </table> A8. ශාක කැප කළ යුතුය/ Crop Cutting Officer's නම/ Name තනතුර/ Designation ලිපිනය/ Address A9. පළමුවෙන් කුඹුර බැලීමට ගිය දිනය Date of first visit to the field } A10. භාවිතා කරන සසම්භාවී පොතේ අංකය Random Book Number } 	විශාල Major	සුළු Minor	වර්ෂා වේෂිත Rainfed	1	2	3	B1. කුඹුරු කැබැල්ලේ නම/අංකය Name of the Parcel/ No. } B2. ගොවි මහතාගේ නම Name of the Farmer } B3. කැබැල්ලේ වැසුරු ප්‍රමාණය Sown extent in Parcel } B4. කැබැල්ලේ ඇති ලියදි ගණන No. of Discernible Square area (Liyadda) in the Parcel } B5. ලියදි තේරීම Selection of Discernible Square area (Liyadda) සසම්භාවී අංක { ලේඛය/ Row පොතේ / in { තීරුව / Column Random Book { අංකය/ Number B6. ලියදි දිග අඩි/ මීටර Length of Discernible Square area Ft./ meters } B7. ලියදි පළල අඩි/ මීටර Breadth of Discernible Square area Liyadda) Ft. meters } B8. අඩි 17/ මීටර් 6 අඩු කර දිග* Length Minus 17 Ft. / 6 meters* } B9. අඩි 17/ මීටර් 6 අඩු කර පළල * Breadth Minus 17 Ft. / 6 meters } B10. බිම් කොටස තේරීම Selection of the Plot සසම්භාවී අංක { ලේඛය/ Row පොතේ / in { තීරුව / Column Random Book { අංකය/ Number
විශාල Major	සුළු Minor	වර්ෂා වේෂිත Rainfed					
1	2	3					

11 කොටස/ PART 11		කේත අංක/ Codes	
උපදෙස්/ Instructions: අදාළ අංකය හෝ අංක කව කරන්න (Encircle the relevant code or codes)		1 කැබැල්ල / Parcel I	11 කැබැල්ල / Parcel II
1. භුක්තිය/ System of Tenure	1.1 තනි අයිතිය (සිත්තක්කර, රජයේ බලපත්‍ර) / Singly Owned (Free hold/ hereditary, Govt. Permit)	1	1
	1.2 හවුල් අයිතිය (තට්ටුමාරු, කට්ටිමාරු ඇතුළුව) / Jointly Owned (including Farmer Rotation & Land Rotation)	2	2
	1.3 අදේ අයිතිය / (Tenant Cultivator).....	3	3
	1.4 අනවසර/ Encroachment.....	4	4
	1.5 වී පොරොන්දු බිම් පංශු, උකස්, කන්න බදු ආදිය/other	5	5
2. කුඹුර සකස් කළ ක්‍රමය / (කේත කිහිපයක් විය හැකිය) Predominant method of preparation of land (Should be Several codes)	2.1 ට්‍රැක්ටරයෙන්/ By Tractor.....	1	1
	2.2 හරකුත් ලවා සෑම/ Buffalo Ploughed.....	2	2
	2.3 හරකුත් ලවා මැඩවීම/ Buffalo Muddled	3	3
	2.4 උදර වලින් කෙටීම/ By Mamotee/ Hoe	4	4
3. වල් නාශක භාවිතය / Usage of Weedicide	3.1 කුඹුරු සකස් කිරීමේදී වල් නාශක භාවිතා කළේද? Did you use weedicide for preparation of land? මව් / Yes	1	1
	නැත / No.....	2	2
4. බිත්තර වී ලබා ගත් මාර්ගය / Source of obtaining Seed Paddy	4.1 තමාගේ / Owned	1	1
	4.2 වෙනත් ගොවියකුගෙන්/ From Other Farmer.....	2	2
	4.3 කෘෂිකර්ම දෙපාර්තමේන්තුවෙන්/ ගොවිජන සංවර්ධන දෙපාර්තමේන්තුවෙන් From Dept. of Agriculture/ Dept. of Agrarian Development.....	3	3
	4.4 වෙනත් (සඳහන් කරන්න) / Other (Specify)	4	4
5. වී වර්ගය (තම සඳහන් කරන්න) / (Variety of Seed Paddy (Specify Name)			
	5.1 රතු කෙටි (රතු සම්බා) / Short Grain (Red)	1	1
	5.2 රතු දිග (රතු නාඩු) / Long Grain (Red)	2	2
	5.3 සුදු කෙටි (සුදු සම්බා) / Short Grain (White)	3	3
	5.4 සුදු දිග (සුදු නාඩු) / Long Grain (White)	4	4
	5.5 වෙනත් (සඳහන් කරන්න) / Other (Specify)	5	5
6. වැසුරු ක්‍රමය / Method of Sowing	6.1 අතින් ඉසීම / Broadcasting	1	1
	6.2 පෙළට සිටුවීම / Transplanted in Rows.....	2	2
	6.3 පෙළට නැතිව සිටුවීම / Transplanted not in Rows.....	3	3
	6.4 පැරෂූට් ක්‍රමයට සිටුවීම / Parachute Method.....	4	4
	6.5 යන්ත්‍ර මගින් සිටුවීම / Planting by Machinery.....	5	5
	6.6 වෙනත් (සඳහන් කරන්න) / Other (Specify).....	6	6

* Where smaller experimental plot sizes are used in terraced fields, the actual length and breadth of these plots should be deducted.

7. පොහොර භාවිතය / Application of Fertilizer	7.1 රසායනික පොහොර පමණක් / Chemical Fertilizer only	1	1
	7.2 කාබනික/ ඓතිහාසික පොහොර පමණක්/ Organic Fertilizer only.....	2	2
	7.3 රසායනික සහ ඓතිහාසික පොහොර යන දෙවර්ගයම / Both Chemical and Organic Fertilizer.....	3	3
	7.4 කිසිත් නැත/ None.....	4	4
	7.A රසායනික පොහොර භාවිතා කර ඇත්නම් ප්‍රමාණය කිලෝ ග්‍රෑම් If Chemical Fertilizer is used, Quantity in killo Grams		
	7.A.1 MOP	kg	kg
	7.A.2 TSP.....	kg	kg
	7.A.3 Urea.....	kg	kg
	7.B කාබනික පොහොර පමණක් භාවිතා කළේ නම්, කාබනික පොහොර පමණක් භාවිතා කළ කණ්ණාඩි ගණන No. of Seasons used If only Organic Fertilizer was Used.		
8. වල් නෙළීම/ Weeding	8.1 අතීන් / Hand Weeding	1	1
	8.2 වල් නාශක මගින් / Using Weedicide	2	2
	8.3 ජලයේ කුණුවීමට හැරීමෙන් / By use of Water	3	3
	8.4 වල් නෙළීමක් නැත / No Weeding	4	4
9. කෘමි නාශක, දිලීර නාශක / Insecticide, Fungicide	9.1 යොදා ඇත / Used	1	1
	9.2 යොදා නැත / Not Used	2	2
10. වගා හානිය / Adverse affects on crop කේත අංක 1,2,3,4,5 කට කළ යුක්ත කැබලිල්ල සම්පූර්ණයෙන්ම පාළු වී ගොවියා එය කපාගෙන නැතත්ම පමණි Codes 1,2,3,4,5 in respect of crop damage should be encircled only if the parcel was severely affected and it was not harvested.	10.1 හානි වී නැත /Not affected	1	1
	10.2 හානි වී ඇති නමුත් කැබලිල්ල කපාගෙන ඇත/ Affected but harvested.....	2	2
	10.3 තදබල ලෙස හානි වී පහත දැක්වෙන හේතු නිසා කැබලිල්ල කපාගෙන නැත Severely affected and the Parcel was not harvested due to following reasons.	3	3
	a. නියතය / Drought.....	1	1
	b. ගංවතුර / Flood	2	2
	c. කෘමීන් නිසා / Pests	3	3
	d. දිලීර රෝග / Fungal diseases	4	4
	e. වෙනත් හේතු/ Other Adverse Factors	5	5
11. ගොයම කපන ලද ක්‍රමය/ Method of Harvesting	11.1 අතීන්/ By manually	1	1
	11.2 ගොයම් කපන යන්ත්‍ර මගින් / By Threshing Machine	2	2
12. ගොයම් මඩින ක්‍රමය/ Method of Threshing	12.1 හරකුන් මගින් / By Buffalo.....	1	1
	12.2 ට්‍රැක්ටර් මගින් / By Tractor.....	2	2
	12.3 ඇග්‍රිමෙක් මගින් / By Agrimec.....	3	3
	12.4 සුනාමි යන්ත්‍ර මගින් / By Tsunami Machine.....	4	4
	12.5 සංයුක්ත අස්වැන්න නෙළීමේ යන්ත්‍ර මගින් / By Combine Harvester.....	5	5
	12.6 පයින් පැහිම හෝ වෙනත් / Manually or Other	6	6
13. පිදුරු භාවිතය / Use of Straw	කුඹුරේ පිදුරු භාවිතා කරන ආකාරය/ Use of Straw in the Parcel		
	13.1 පොහොර වශයෙන් කුඹුරට යෙදීම / Apply to the Field as Fertilizer.....	1	1
	13.2 ගිනි තැබීම / Firing.....	2	2
	13.3 සත්ව ආහාර සඳහා / For Animal Feed	3	3
	13.4 වහල සෙවිලි කිරීම සඳහා / For Roofing.....	4	4
	13.5 වෙනත් /Other.....	5	5

14. 111 කොටස/ PART 111 ශාක නෙළීමේ ප්‍රතිඵල / Results of Crop Cutting Experiments			
14. i. අස්වැන්න / Yield	ප්‍රගණන ලකුණු වලින් දක්වන්න / Enter in tally marks	1 කොටස/ Plot I	11 කොටස/ Plot II
14.ii. සම්පූර්ණ ප්‍රමාණය (ඉලක්කමෙන්) / Total Volume (in Number)			
14.iii. කාර්යාල ප්‍රයෝජනය සඳහා / For office use only			
14.iv. බිම් කොටසේ ප්‍රමාණය දක්වමින් රූ සටහනක් අඳින්න / Draw a Sketch diagram of the plot giving dimensions			
		1 කැබැල්ල / Parcel I	11 කැබැල්ල/ Parcel II
14.v. කුඹුරු කැබැල්ලේ අපේක්ෂිත අස්වැන්න / Expected Yield in the Parcel	බුසල් / Bushels		
14.vi. මෙම කණ්ණාඩියේ වගා කළ සියලුම කුඹුරු කැබලිවලින් ඔබට ලැබුණු වී අස්වැන්න යොදා ගැනීමට බලාපොරොත්තු වන ආකාරය/ What do you intend to do for your share of yield obtained from all parcels cultivated in this season			
a. කෑම සඳහා පමණක් / For Home consumption only	1	1	
b. විකිණීම සඳහා පමණක් / For Sale only	2	2	
c. කෑම සහ විකිණීම සඳහා / For Home consumption and Sale.....	3	3	
d. වෙනත් (සඳහන් කරන්න) / Other (Specify)	4	4	
14.vii. ශාක නෙළ දිනය / Date of harvesting	අවු/ Yr Mo./මා Da./දි 20	අවු/ Yr Mo./මා Da./දි 20	

1 කැබැල්ල / Parcel I

11 කැබැල්ල/ Parcel II

ගොවි මහතාගේ අත්සන / Signature of Farmer (Cultivator)

ගොවි මහතාගේ දුරකතන අංකය / Telephone No. of Farmer (Cultivator)

ශාක නෙළන නිලධාරියාගේ / Signature of Crop Cutting Officer

ජනලේඛන හා සංඛ්‍යාලේඛන දෙපාර්තමේන්තුව
DEPARTMENT OF CENSUS AND STATISTICS

වී අස්වැන්න ඇස්තමේන්තු කිරීමේ සමීක්ෂණය } මහ / Maha
Crop Estimating Survey on Paddy } යල / Yala
වර්ග පරීක්ෂණ නිලධාරියාගේ වාර්තාව/Report of Quality Checking Officer

A. දිස්ත්‍රික්කය District
B. ප්‍රා. ලේ/ප්‍රා. උ. දි. කොට්ඨාස D. S./A. G. A. Div.
C. ප්‍රා. නි. කොට්ඨාස G. N. Div.
D. ගම් Village

E. ගණන තෙලන නිලධාරියාගේ නම සහ තනතුර }
Name and Designation of Crop Cutting Officer

1. මිම්ම කොටස් මැනුම් පරීක්ෂණ කරන ලදී }
Measurements of the Plot were checked
2. ගණන තෙලන නිලධාරියා විසින්, ජන හා සං. දෙ. විසින් සපයා ඇති ස්ථරු කඩ්වල් භාවිතා කරන ලද්දේ ද? }
Did the Crop Cutting Officer use the standard Measuring Sets provided by D. C. S. ?
3. සම්පූර්ණ අස්වැන්න ප්‍රමාණය }
Total Volume of Paddy }
ස්ථරු/Seers
ස්ථරු 1/32 1/32/Seers
4. පරීක්ෂණය සඳහා ගතවූ කාලය-වැය }
Time taken for the Experiment-hours

1 කොටස/plot 1	2 කොටස/plot 2
සරි/වැරදි* Correct/Not correct*	සරි/වැරදි* Correct/Not correct*
මිටි/තැන්*	මිටි/තැන්*

(අ. 8. බ/ප.ට.ඔ)

5. මෙම සමීක්ෂණය, නිසි අයුරින්/තහන දැක්වෙන දේ සහිතව සිදු කළේ නැත නිසා සමීක්ෂණය කරමි.
I am of the view that the experiments were properly conducted/not conducted due to reasons given below*

.....

වර්ග පරීක්ෂණ නිලධාරියාගේ—
Quality Checking Officer's
නම }
Name }
තනතුර }
Designation }
අත්සන }
Signature }
දිනය }
Date }

*අනවශ්‍ය දේ ඔත්ත දමන්න/*Strike off which is not applicable.

භූමි රේඛිත, බදුල්ල, මහනුවර, කුරුමිල සහ රත්නපුර (අනුමත ප්‍රා. උ. දි. කොට්ඨාස 5) දිස්ත්‍රික්කයන්හි කුඩා ලියවිල 16 1/2' * 16 1/2' (මී. 5 සෙ. මී. 3) සම්මත මිම්ම කොටස යොදාගත නොහැකි වීම පහත සඳහන් මිම්ම කොටස් යොදා ගත යුතුය.
In terraced fields of Nuwara-Eliya, Badulla, Kandy, Kegalle and Ratnapura (in 5 approved A. G. A. Divisions) Districts the following plot sizes are adopted where the standard 16 1/2' * 16 1/2' (5m 3cm) size cannot be laid.

24'9"x11'	(මී. 7 සෙ. මී. 54 x මී. 3 සෙ. මී. 35)
16'6"x8'3"	(මී. 5 සෙ. මී. 3 x මී. 2 සෙ. මී. 51)
12'4 1/2"x5'6"	(මී. 3 සෙ. මී. 77 x මී. 1 සෙ. මී. 68)
4'x3'	(මී. 1 සෙ. මී. 22 x සෙ. මී. 91)

මෙම ආකෘති පත්‍රයෙන් පිටපත් දෙකක් සකස් කර ගණන තෙලීම කරනු ලැබූ දින, එකක් ජන ලේඛන හා සංඛ්‍යාලේඛන අධ්‍යක්ෂ ජනරාල්, කැ. පෙ. 563, කොළඹ 07, යන ලිපිනයට ද අනෙක දිස්ත්‍රික් ලේකම්ට ද යවන්න.

Two copies of this form should be prepared and posted on the date of harvesting one to the Director General of Census and Statistics, P. O. Box 563, Colombo 07, and the other to the D. S.

ජනලේඛන හා සංඛ්‍යාලේඛන දෙපාර්තමේන්තුව
DEPARTMENT OF CENSUS AND STATISTICS

වී අස්වැන්න ඇස්තමේන්තු කිරීමේ සමීක්ෂණය
Crop Estimating Survey on Paddy

මහ / Maha
යල / Yala

කම්කරු ගෙවීම් ලඳුපත / Labor Payment receiving receipt

A. දිස්ත්‍රික්කය
District

B. ප්‍රා.ලේ./ප්‍රා.උ.දි.කො.
D.S. / A.G.A. Div.

C. ග්‍රා.නි.වසම
G.N. Div.

D. ගම
Village

E. වී අස්වැනු සමීක්ෂණයට සහාය වීම සඳහා කම්කරු කුලී ලෙස රු..... ක මුදලක් භාරගනිමි. / Received sum ofRs. to assist the paddy crop estimating survey.

F. අස්වැනු සමීක්ෂණ කටයුතු සඳහා යොදාගත් කම්කරුවාගේ නම සහ අත්සන
Name and Signature of the Laborer

}
.....

G. නියමිත පරිදි අදාළ ගමෙහි අස්වැනු සමීක්ෂණ කටයුතු සිදුකරන ලදි
The experiment were properly conducted

ගසා නිලධාරියාගේ / Crop cutting officer

නම / Name

තනතුර / Designation

අත්සන / Signature

දිනය / Date

}
.....
.....
.....

Table A1 : Seed Variety with Average Life Cycle (Harvesting Period)

	Seed Variety	Dates	GrainType		Seed Variety	Dates	GrainType	
1	AT 303	90	2	49	BG 455	135	2	GrainType
2	AT 306	90	4	50	BG 745	150	3	1 Short Grain (Red)
3	AT 307	90	4	51	BG 750	75		2 Long Grain (Red)
4	AT 308	90	4	52	BG 90-2	120		3 Short Grain (White)
5	AT 309	90	4	53	BG 94-1	105	4	4 Long Grain (White)
6	AT 311	90	2	54	BG 94-2	105		
7	AT 353	105	2	55	BW 372	105		
8	AT 354	105	4	56	BW 267-3	105	4	
9	AT 362	105	2	57	BW 272-6B	90	1	
10	AT 373	105		58	BW 302	90		
11	AT 401	135	2	59	BW 351	105	2	
12	AT 402	135	2	60	BW 361	105	2	
13	AT 405	135		61	BW 363	105	4	
14	Attakari	105	2	62	BW 364	105	2	
15	Batapolaal	105		63	BW 367	105	3	
16	BG 38	150	3	64	BW 372	105	2	
17	BG 205	75		65	BW 400	135	2	
18	BG 250	75	4	66	Dickwee	105		
19	BG 251	75	4	67	Gonabaru	150	4	
20	BG 252	75	1	68	H4	135	2	
21	BG 300	90	4	69	Hodarawalu	135		
22	BG 301	90		70	Hathiyal	150	2	
23	BG 304	90	4	71	Kahamala	105		
24	BG 305	90	4	72	Kahata wee	105		
25	BG 310	90	4	73	Kahawanu	105	3	
26	BG 34-6	105	2	74	Kaluheenati	105	2	
27	BG 34-8	90		75	Kuruluthuda	135	2	
28	BG 3-5	150	4	76	LD 253	75	2	
29	BG 350	105	2	77	LD 355	105		
30	BG 351	105		78	LD 356	105	1	
31	BG 352	105	4	79	LD 365	105	1	
32	BG 357	105	4	80	LD 368	105	1	
33	BG 358	105	3	81	LD 371	105	3	
34	BG 359	105	4	82	LD 408	135	2	
35	BG 360	105	3	83	Madathawalu	105	2	
36	BG 366	105	4	84	MahaMawee	150		
37	BG 367	105		85	Mawee (PS)	150	2	
38	BG 369	105	4	86	MottaiKaruppan	105	2	
39	BG 370	105	3	87	Murungakayan	135	2	
40	BG 374	105	4	88	Pachchaperumal	90	2	
41	BG 379/2	120	4	89	Pokkali	105		
42	BG 380	135	4	90	pokuru Samba	135	3	
43	BG 400-1	135	4	91	Raththal	150	3	
44	BG 403 (Mahasen)	120	4	92	Red Keeri	105		
45	BG 406	120	2	93	Suduheenati	135	4	
46	BG 407 H	150	4	94	Suduru Samba	105	3	
47	BG 450	135	3	95	Suandal	105	3	
48	BG 454	135						

